Counting Divisors

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1713 | Mathematics | https://cses.fi/problemset/task/1713

Given n integers, your task is to report for each integer the number of its divisors.

For example, if x=18, the correct answer is 6 because its divisors are 1,2,3,6,9,18.

Input

The first input line has an integer n: the number of integers.

After this, there are n lines, each containing an integer x.

Output

For each integer, print the number of its divisors.

Constraints

- $1 \le n \le 10^5$
- $1 \le x \le 10^6$

Example

Input
3 16 17 18

Output
5 2 6